

Bio 152 | Spring 2025

Independent Project First Draft Self-evaluation

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Collaborator(s) if applicable:

In thinking about your First Draft, first consider the guiding questions below. Then under each section listed provide a few sentences pertaining to that element, with an honest evaluation of how you are feeling about that aspect of your work. You do not have to answer each question individually. They are just prompts to help guide your reflection.

Final paper grades will be based on the clarity, logic, and depth of your work. This self-evaluation is meant to help you think through where you should focus your efforts in the next submission. This worksheet also gives your IP reviewer insight into your own thinking so they know better how to help guide you to successful project completion.

1. TITLE AND ABSTRACT (Not required for the First Draft)

- a. Does the title reflect your study's purpose?
- b. Does the title reflect your preliminary results or interpretation of them?
- c. Do you summarize your investigation in a brief paragraph Abstract (~120 words)?
- d. In your abstract, if applicable, is it clear that this is based on preliminary research?
- e. In your abstract, do you include your purpose, rationale, brief design (methods), results, and conclusions?

Which of the criteria listed above went very well, and what parts do you think you could use some help with? Providing specifics will help you and your reviewer. Add space as you need to.

- a. The title will need to be updated as right now it does reflect but it simply isn't expressive of the goals
- b. No it is too general
- c. No we have the space but I wanted to wait till I write the analysis so I can include a description in the abstract

2. INTRODUCTION

- a. Is there a clear statement of purpose for the research?
- b. Is the biological rationale for the research clearly stated and logically connected with an explanation of the biological mechanisms involved? Is background information from the literature presented to support this rationale?
- c. Is the broader justification for the study (impact to science and humanity) clearly stated and logically connected? Is background information from the literature presented to support this justification?
- d. Is your testable hypothesis clearly stated?

Which of the criteria listed above went very well, and what parts do you think you could use some help with? Providing specifics will help you and your reviewer. Add space as you need to.

- a. Yes there is
- b. Yes we do talk about that in the first section reference the Mayo clinic and one other
- c. I think we could do a little more about its impact. It shouldn't be difficult to add however, we can state it more directly. Yes there is background support

d. *Yes we do state it directly.*

3. METHODS

- a. Do your methods test your stated hypothesis?
- b. Do you outline how you will collect or did collect the data? When appropriate, is there an explanation of how others collected and compiled your data?
- c. Is there a clear description of and justification for the methods used to analyze the data?
- d. Is your description specific enough that a reader can understand exactly what you plan to do or did and replicate it?

Which of the criteria listed above went very well, and what parts do you think you could use some help with? Providing specifics will help you and your reviewer. Add space as you need to.

- a. *Yes for each subsection we talk about what we test*
- b. *Yes we mention how the database was collected and how we grabbed that data and any flaws*
- c. *Yes I describe why the code was the way it was and provide figures of it*
- d. *It is hard to determine that as pandas and numpy are basic python libraries but to the audience in this class perhaps not but we will mention that they were used*

I think I should go back and add in some explanation on what numpy, pandas, and matlab.plot are and what we use them for. Although they are very common in the scientific field it wont hurt to add some explanation.

4. RESULTS & DISCUSSION (abbreviated for the First Draft)

- a. Did you describe the appropriate statistical test(s) for your anticipated data?
- b. Is the figure, table, image, or other visual representation of your data you plan to use an ideal way to help the reader understand your analyses and results?
- c. Are the interpretations of your potential results reasonable for the type of data you (will) have?
- d. Did you describe the connection of your potential conclusions to the overall purpose of your research and the broader implications of your study as described in your Introduction?
- e. Are possible limitations and alternative explanations discussed?

Which of the criteria listed above went very well, and what parts do you think you could use some help with? Providing specifics will help you and your reviewer. Add space as you need to.

- a. *I believe we used the appropriate tests*
- b. *Yes the plots are good but I need to make proper figures instead of using the code comments for the other data*
- c. *Yes they are*
- d. *That section still needs to be finished as again I want to fully complete my analysis*
- e. *We need to complete it but the section was there*

5. REFERENCES

- a. Do you cite any sources you've used throughout all sections of your paper?
- b. Are all non-original claims supported by literature?
- c. Are the references cited correctly?

*Which of the criteria listed above went very well, and what parts do you think you could use some help with?
Providing specifics will help you and your reviewer. Add space as you need to.*

- a. Yes we do however we need to organize them properly
- b. Yes
- c. No we need to organize them in order properly

6. OVERALL WRITING QUALITY

- a. Is the First Draft well organized?
- b. Does it flow logically?
- c. Do you use transitions when appropriate (e.g. between sections to help tie them together)?
- d. Do you use proper punctuation, grammar, and spelling?
- e. Do you use the same tense throughout the paper?
- f. Is the paper clear, concise, and readable?

*Which of the criteria listed above went very well, and what parts do you think you could use some help with?
Providing specifics will help you and your reviewer. Add space as you need to.*

- a. Yes I believe it is structure properly
- b. I believe it does flow logically
- c. The sections help the reader understand
- d. As far as I could tell
- e. This needs to reviewed however once we write the analysis it shouldn't be too hard
- f. I believe so.

7. LEARNING PROCESS & RESOURCE USE

- a. What aspects of your project do you understand better after completing the First Draft?
- b. What do you hope to understand better before submitting your Second Draft? Are there things you plan to change about your approach to writing for the next submission?
- c. Did you use any AI resources to gather information, find data, brainstorm topics, or help with other pre-writing tasks? *(You will not be penalized for using AI in this manner. Remember, AI may not be used to write any aspect of your IP submissions).*

- a. I believe the project is pretty self-explanatory but perhaps I am missing something
- b. I believe the one section I plan to change is flipping the methods section to first describe the process then the data sets but outside of that no.

- c. I used chatgpt to keep track of my sources and format them as I went but it always does a poor job so its mostly just to archive and quickly get a basic form down

Lastly, if you had to grade your paper on a scale of 1 to 5, how would you score it? Use the criteria below.

SCORE: 3

Because we still have some things to fix up that were left unedited however the parts we did were done well

- Poor = 1
 - This paper has fundamental flaws. It might be hard to understand, be completely off topic, or be missing essential components. While some of the required elements might be present in the draft, none of them were successful.
- Fair = 2
 - This submission has some substantial issues. There might be the beginnings of good elements, but they are far from complete or clear.
- Good = 3
 - This submission is on the right track but might be lacking detail and/or content depth. It might take some effort to understand the study and biological rationales for the project.
- Very good = 4
 - The paper is successful in conveying information clearly and there is logical flow throughout the paper. There are still some small improvements to be made, but it is in very good shape at this stage.
- Excellent = 5
 - All the elements are executed extremely well. The content is easy to understand, and its importance is clear. Any suggested changes are purely a matter of reader preference and don't impact clarity.